

Title

Individualized, risk-stratified screening intervals for diabetic retinopathy: design and rationale of a large-scale population equivalence trial

Abstract

Rationale: Annual screening for diabetic retinopathy imposes substantial resource burden, yet individualized risk assessment might optimize screening intervals while maintaining safety. This equivalence trial evaluates a risk-stratified, variable-interval screening approach versus standard annual screening. **Methods:** We will enroll 4500+ individuals with diabetes aged 12+ years from an established population-based screening program. Participants will be randomized 1:1 to receive either individualized screening intervals (6, 12, or 24 months based on baseline and dynamic risk factors) determined by a validated risk prediction algorithm, or annual screening (control). Screening intervals will be adjusted at each visit based on updated risk calculations incorporating demographic, clinical, and prior screening data. Primary outcome is first-visit attendance (safety measure). Secondary outcomes include detection rates of sight-threatening retinopathy, quality of life measures, and program costs. The study will employ a two-year minimum follow-up with observer masking of allocation. The equivalence margin for attendance is pre-specified.